

OSSP (Operating Systems And Systems Programming) - 25CS2104E

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SEC-8

SKILL-9:

Aim: To Change Directories, Validate Paths, Update Working Directory, Handle Errors, Maintain Previous Directory, Test Navigation.

To Identify Built-ins, Create Dispatch Table, Execute In-Process Commands, Handle Invalid Commands, Maintain State, Test Dispatch Logic.

PROCEDURE :

1. Change Directory :

```
siddinapujith07@DESKTOP-FDHCAIT:~$ cd skill9
siddinapujith07@DESKTOP-FDHCAIT:~/skill9$ nano cd_program.c
siddinapujith07@DESKTOP-FDHCAIT:~/skill9$ gcc cd_program.c -o cd_program
siddinapujith07@DESKTOP-FDHCAIT:~/skill9$ ./cd_program
Current directory: /home/siddinapujith07/skill9
Enter directory path: /tmp
Directory changed successfully.
New directory: /tmp
Previous directory: /home/siddinapujith07/skill9
siddinapujith07@DESKTOP-FDHCAIT:~/skill9$ ./cd_program
Current directory: /home/siddinapujith07/skill9
Enter directory path: /abc123
Error changing directory: No such file or directory
```

2. Built-in Command Dispatcher :

```
siddinapujith07@DESKTOP-FDHCAIT:~/skill9$ nano builtin.c
siddinapujith07@DESKTOP-FDHCAIT:~/skill9$ gcc builtin.c -o builtin
siddinapujith07@DESKTOP-FDHCAIT:~/skill9$ ./builtin
Simple Built-in Shell
shell> pwd
/home/siddinapujith07/skill9
shell> cd /tmp
shell> pwd
/tmp
shell> cd /home
shell> pwd
/home
shell> hello
Invalid command: hello
shell> exit
Exiting shell...
```

Code :

```
GNU nano 8.7.1 cd_program.c *
#include <stdio.h>
#include <unistd.h>
#include <string.h>
#include <stdlib.h>
#include <limits.h>

int main()
{
    char path[PATH_MAX];
    char previous[PATH_MAX];
    char current[PATH_MAX];

    getcwd(current, sizeof(current));
    strcpy(previous, current);

    printf("Current directory: %s\n", current);
    printf("Enter directory path: ");
    scanf("%s", path);

    if (chdir(path) == 0)
    {
        strcpy(previous, current);
        getcwd(current, sizeof(current));

        printf("Directory changed successfully.\n");
        printf("New directory: %s\n", current);
        printf("Previous directory: %s\n", previous);
    }
    else
    {
        perror("Error changing directory");
    }

    return 0;
}
```

```
#include <stdio.h>
#include <string.h>
#include <unistd.h>
#include <stdlib.h>

void pwd_command()
{
    char path[1024];

    if (getcwd(path, sizeof(path)) != NULL)
        printf("%s\n", path);
}

void cd_command(char *path)
{
    if (chdir(path) != 0)
        perror("cd");
}

void exit_command()
{
    printf("Exiting shell...\n");
    exit(0);
}

int main()
{
    char command[100];
    char *argument;

    printf("Simple Built-in Shell\n");

    while (1)
    {
        printf("shell> ");
        fgets(command, sizeof(command), stdin);
```

```

        command[strcspn(command, "\n")] = '\0';

        argument = strtok(command, " ");

        if (argument == NULL)
            continue;

        if (strcmp(argument, "pwd") == 0)
        {
            pwd_command();
        }
        else if (strcmp(argument, "cd") == 0)
        {
            argument = strtok(NULL, " ");

            if (argument != NULL)
                cd_command(argument);
            else
                printf("Usage: cd <directory>\n");
        }
        else if (strcmp(argument, "exit") == 0)
        {
            exit_command();
        }
        else
        {
            printf("Invalid command: %s\n", argument);
        }
    }

    return 0;
}

```

Observation :

The programs successfully changed and validated directory paths, displayed the current and previous directories, handled invalid paths, and executed built-in commands such as cd, pwd, and exit. Invalid commands were also handled with appropriate error messages.

Result :

The Change Directory and Built-in Command Dispatcher programs were successfully implemented and executed, demonstrating directory navigation, path validation, error handling, state maintenance, and built-in command execution.